Water storage change across Thailand

Climate memory, ENSO and attribution credibility

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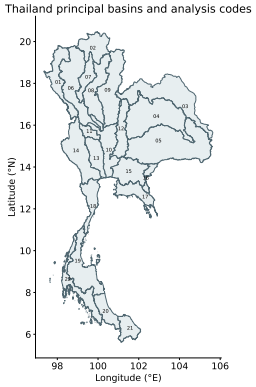
Research focus

- How are total and groundwater storage changing across Thailand’s 22 principal basins?
- How much of the observed change can the climate baseline explain?
- How strongly do independent observations support the remaining residual?

Research objectives

Objective	Output
1 Storage change	Monthly TWSA/GWSA, Sen slopes and Mann–Kendall tests
2 Climate and residual	Climate reconstruction, GLDAS benchmark and observation checks
3 Decision support	Interactive basin app, input data and reproducible code

Thailand’s 22 principal basins



- Official ONWR reporting framework
- GRACE information extends across basin boundaries
- All basin tables retain coverage and uncertainty

ONWR principal-basin geometry. Reporting units do not imply independent GRACE observations.

Analysis workflow

- Harmonize monthly units, dates, area weights and the 2004–2009 reference
- Calculate GWSA and estimate storage trends
- Reconstruct naturalized storage using weather and climate memory
- Calculate the residual, examine uncertainty and compare independent observations

Groundwater storage balance

$$\mathrm{GWSA} = \mathrm{TWSA} - \mathrm{SMSA} - \mathrm{SWSA} - \mathrm{CWSA} - \mathrm{SWEA}$$

- Subtract soil, surface, canopy and snow storage anomalies
- Use common valid cells and a common reference period
- Groundwater uncertainty includes every subtracted compartment

Storage anomalies use mm of equivalent water depth. Snow is small in Thailand but retained consistently.

Residual accounting

$$R = O - C = \underbrace{(A - C)}_{H} + \underbrace{(O - A)}_{E}$$

- H describes the human effect represented by the paired model
- E retains the observation–historical-model discrepancy
- The primary residual uses the published naturalized baseline

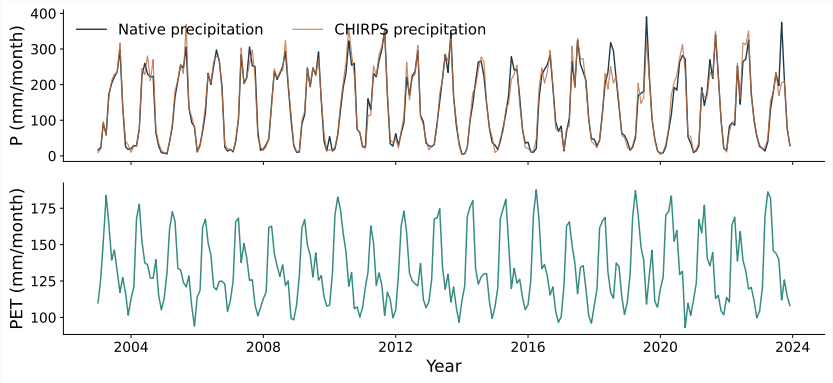
The statistical emulator reconstructs C from weather. Its prediction does not replace C in the primary residual.

Chronological model evaluation

Period	Years	Use
Warm-up	1981–1990	Initialize climate memory
Training	1991–2009	Fit transformations and tune models
Validation	2010–2015	Freeze model selection
Test	2016–2023	Evaluate unseen years

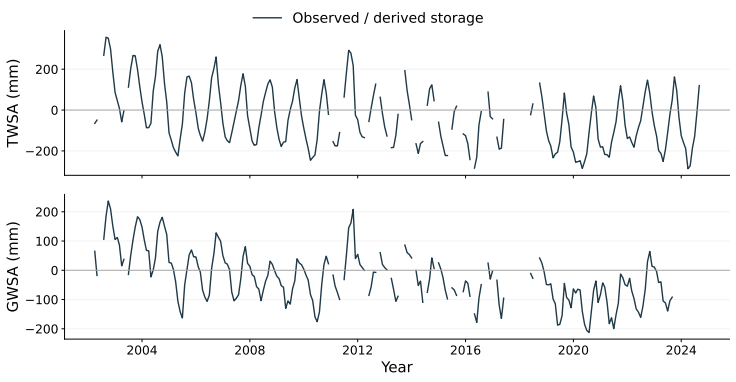
Ridge, Elastic Net, splines and Random Forest. Predictors exclude GRACE and groundwater observations.

Precipitation and potential ET



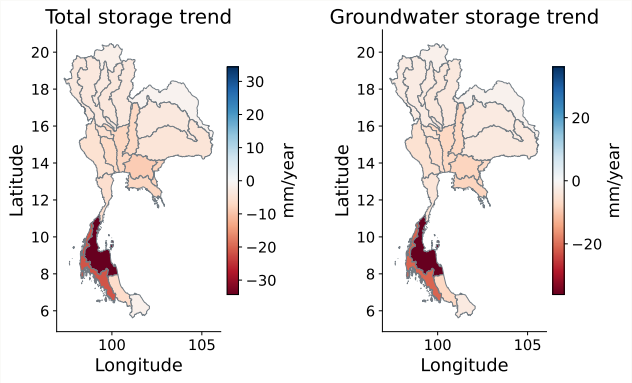
National monthly values, 2003–2023. Native P/PET drive reconstruction. CHIRPS supplies the rainfall replacement test.

Observed storage through time



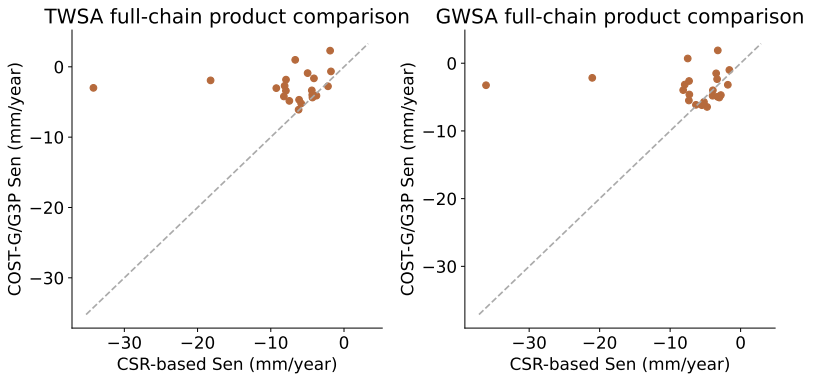
National CSR TWSA and compartment-derived GWSA. Missing months remain gaps. Groundwater ends earlier because of compartment availability.

Storage trend patterns across basins



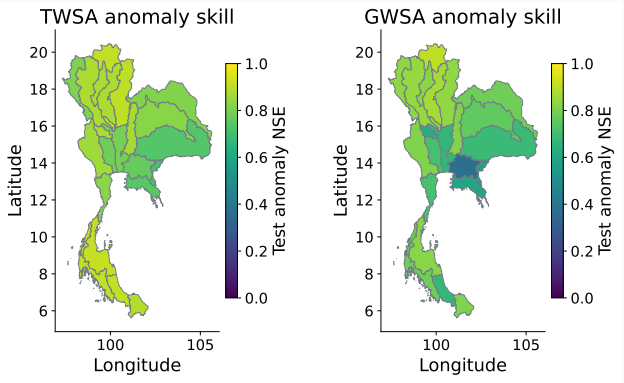
Seasonal Sen slopes. Negative values indicate decline. All basins retain the CSR small-area caution.

Basin ranking depends on the processing chain



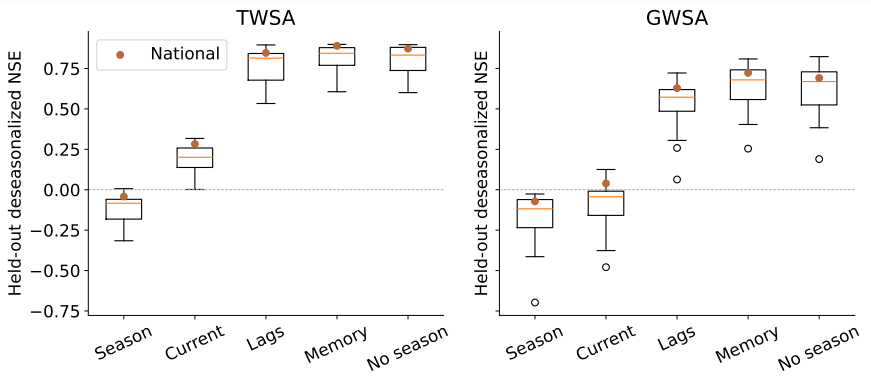
Common-date rank correlations: TWSA 0.138 and GWSA 0.027. Groundwater chains also differ in compartment processing.

Climate reconstruction performance



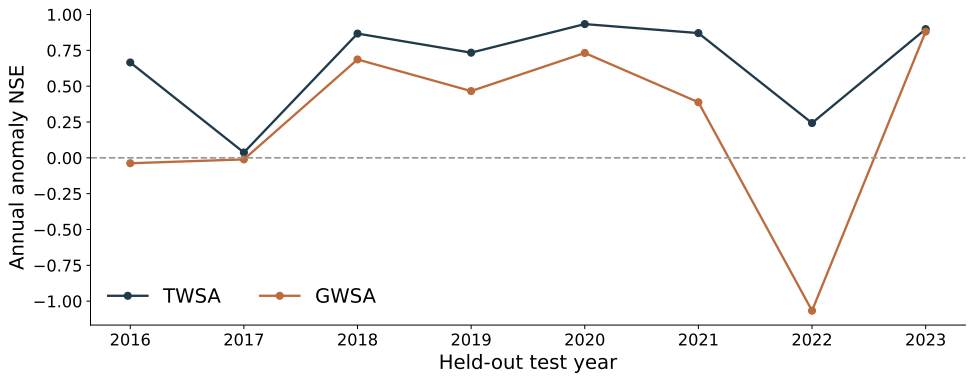
Ensemble test anomaly NSE against naturalized WaterGAP, 2016–2023. High scores verify reconstruction of the parent model.

Climate memory improves reconstruction



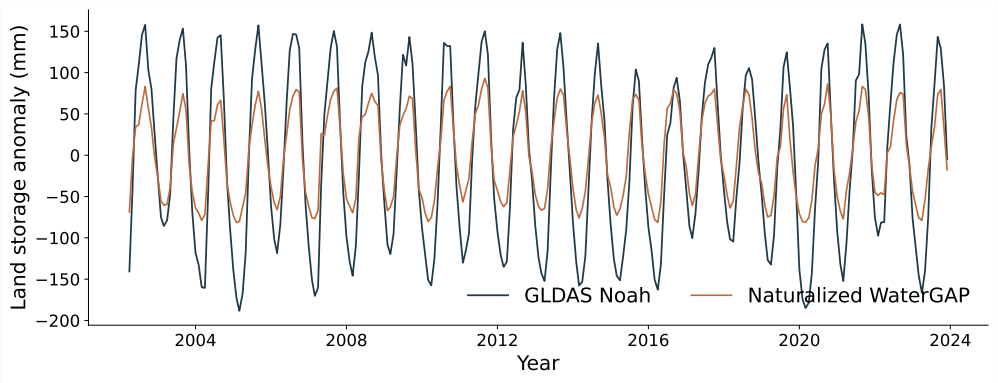
Controlled Ridge experiment. National test anomaly NSE: current weather / memory = 0.283 / 0.890 for TWSA and 0.039 / 0.723 for GWSA.

Performance varies between test years



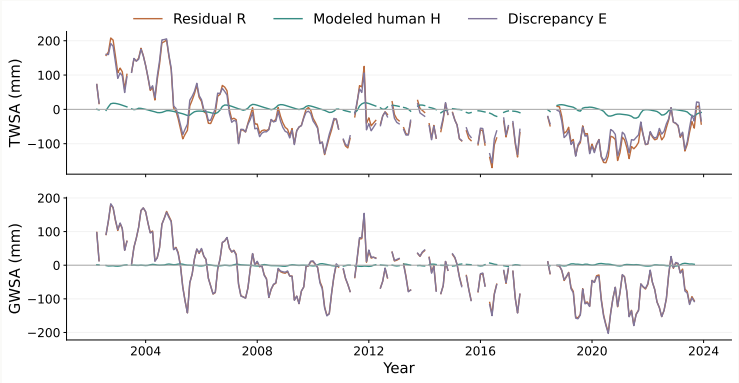
2022 national anomaly NSE: TWSA 0.234; GWSA −1.067. Negative NSE indicates performance below the mean benchmark.

GLDAS comparison on matching compartments



Soil + canopy + snow only. National r = 0.969, NSE = 0.738, RMSE = 52.49 mm across 261 months. Groundwater and surface water are excluded.

Residual and modeled human effects differ



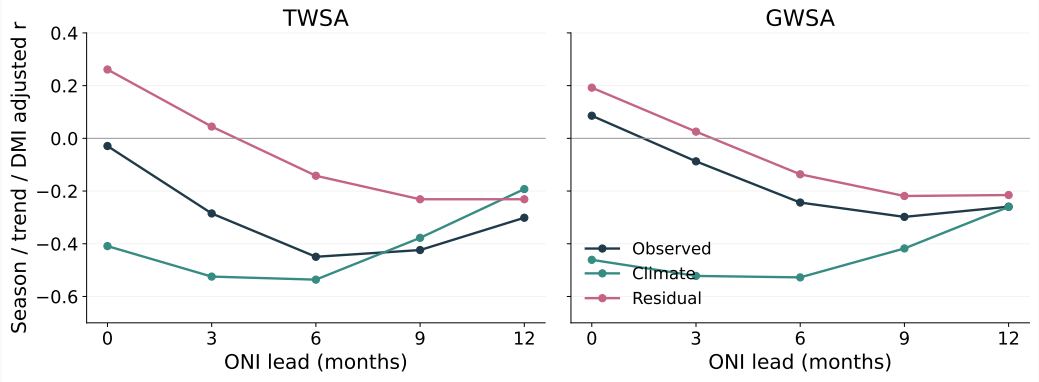
R = H + E on common observed months. The remaining discrepancy includes unresolved model and observation errors.

Additive national trend partition

Component	TWSA	GWSA
Observed O	-8.36	-7.28
Naturalized C	-0.54	-0.21
Modeled human H	-0.49	0.06
Residual R	-7.81	-7.07
Discrepancy E	-7.32	-7.13

OLS slopes, mm/year, using the same months. R = H + E. Seasonal Sen slopes are reported separately because their medians are not additive.

ENSO is clearer in the climate baseline than in the residual



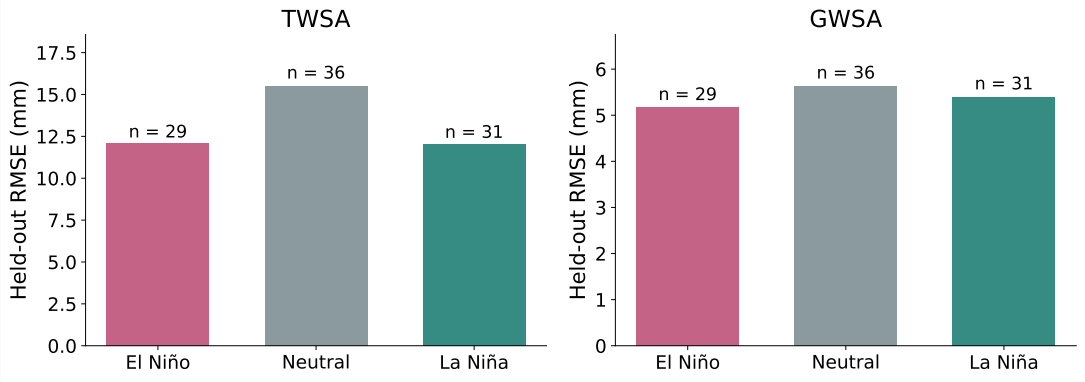
National 2003–2023 diagnostics. At six months, TWSA baseline r = −0.536 and residual r = −0.142; GWSA values are −0.528 and −0.136. These differences are descriptive, not a causal validation.

ENSO evidence remains limited after multiple comparisons

- All five selected leads and both ONI and historical RONI v5 are retained
- No basin association passes the complete FDR screen
- Only 21 years: minimum whole-year rotation p = 1/21
- Weak residual correlation does not prove complete climate removal

Circular whole-year rotations preserve index patterns but assume shift comparability. Contemporary DMI, calendar season and trend are controlled. Current operational RONI uses v6; the study freezes historical v5.

Reconstruction skill varies with ENSO phase



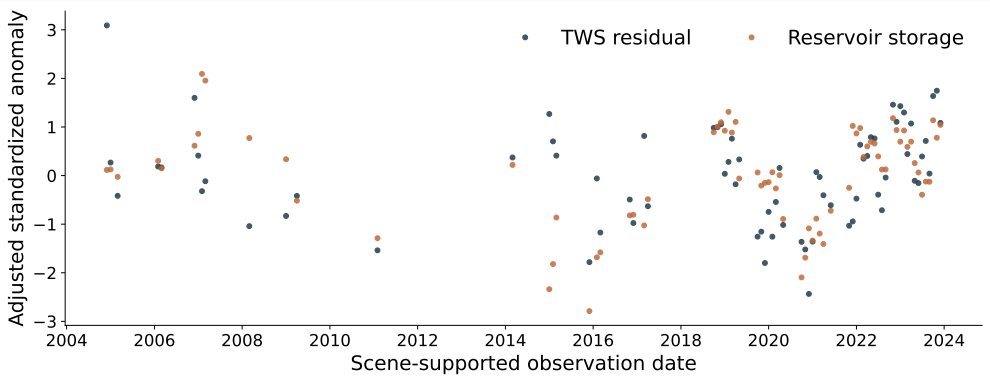
Frozen 2016–2023 ensemble, naturalized parent target. There are 29 El Niño, 31 La Niña and 36 neutral months. The months share events; these are conditional reconstruction scores, not independent event forecasts.

Well selection changes residual agreement

Station panel	TWS r	GWS r	GWS years
Longest overlap	0.127	0.037	17
Fixed eligibility / wider history	0.657	0.511	17
Common period	0.325	0.029	8
Strictly balanced	0.305	≈0.000	8

511 wells provide 5,484 aggregates. Co-located wells are grouped. The common-period panels cover eight usable storage years within 2013–2022.

Independent reservoir evidence



26-reservoir fixed panel, 73 scene-supported months. Adjusted r = 0.439, year-cluster 95% interval 0.188–0.665. Both curves remove season and linear trend.

Decision support and reproducibility

- Interactive storage histories, trend maps and observation support for all 22 basins
- Downloadable basin performance tables and spatial/temporal forcing figures
- Input subsets, pinned dependencies and executable scientific workflows

Interactive basin analysis, a consolidated figure library, final documents and runnable code:

https://pratikkhanal16.github.io/thailand-water-storage-dss/

Conclusions

- National total and derived groundwater storage decline in the primary records
- Weather memory improves parent-model reconstruction; ENSO diagnostics add context
- Processing chains and monitoring support limit local rankings and attribution
- The residual remains human effects plus unresolved discrepancy

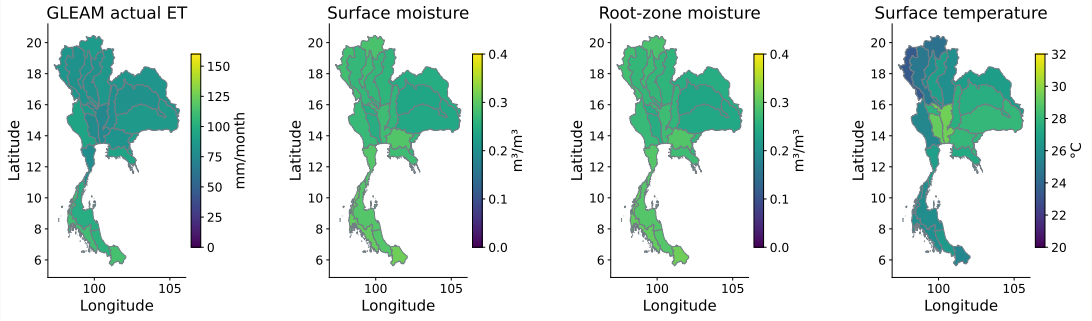
Contribution: a reproducible joint assessment of product sensitivity, climate memory, paired-model discrepancy and monitoring-network uncertainty.

Selected references

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- Müller Schmied et al. (2024), WaterGAP. doi:10.5194/gmd-17-8817-2024
- Chueasa et al. (2024), ENSO and Thai rainfall. doi:10.1016/j.mex.2024.102954
- Sharifi et al. (2025), spatial response. doi:10.5194/hess-29-6985-2025
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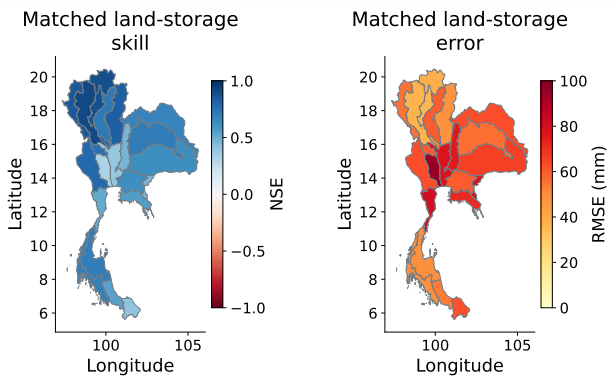
Metropolis theme: Matthias Vogelgesang and contributors, https://github.com/matze/mtheme. Full source bibliography and frozen NOAA indices accompany the report.

Spatial hydroclimatic context



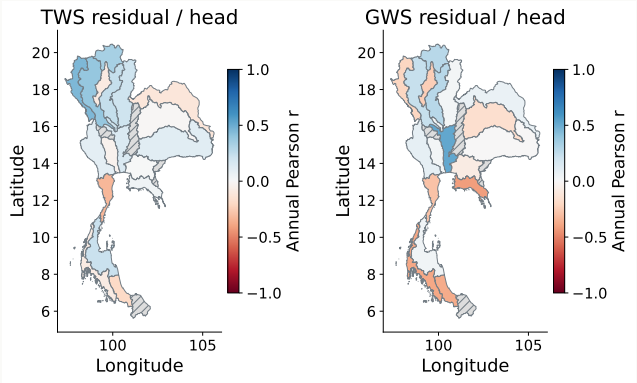
Basin available-month means, 2003–2023. Product coverage differs. Volumetric moisture is distinct from storage in mm.

GLDAS agreement varies among basins



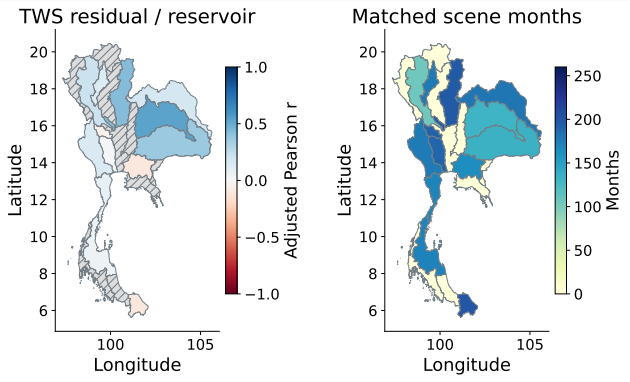
Matched land-storage benchmark. NSE colours saturate below −1 and RMSE above 100 mm. GLDAS also contributes to the groundwater subtraction.

Spatial groundwater consistency



Residual–head correlations on available annual pairs. Grey hatching marks insufficient evidence. These comparisons cannot validate a human-only total.

Reservoir agreement and observation support



Scene-screened MSEA-Res Level 1. Reservoir storage checks a surface-water component. Climate and operating decisions both affect reservoirs.

TWSA climate reconstruction performance (01–11)

Basin	n	r	NSE	Anom NSE	RMSE	MAE
01	96	0.991	0.981	0.807	22.08	16.90
02	96	0.990	0.979	0.904	17.32	13.88
03	96	0.990	0.973	0.812	27.37	19.39
04	96	0.981	0.961	0.821	15.61	11.58
05	96	0.974	0.946	0.732	21.43	15.40
06	96	0.991	0.981	0.876	15.90	12.00
07	96	0.987	0.974	0.868	15.24	11.08
08	96	0.987	0.972	0.873	15.38	11.43
09	96	0.992	0.983	0.901	14.14	10.83
10	96	0.967	0.928	0.792	42.23	30.13
11	96	0.990	0.980	0.875	15.34	12.27

Frozen ensemble, 2016–2023 test data. Target: naturalized WaterGAP, not GRACE. Errors are mm; anomaly NSE removes the training monthly climatology. Higher NSE and lower errors are better.

TWSA climate reconstruction performance (12–22)

Basin	n	r	NSE	Anom NSE	RMSE	MAE
12	96	0.987	0.974	0.864	11.22	8.35
13	96	0.974	0.942	0.778	32.00	23.99
14	96	0.991	0.981	0.856	18.49	14.13
15	96	0.975	0.942	0.764	28.65	22.72
16	96	0.979	0.957	0.705	14.60	11.45
17	96	0.987	0.972	0.742	16.75	13.62
18	96	0.979	0.957	0.827	14.38	10.97
19	96	0.982	0.965	0.911	17.38	13.55
20	96	0.982	0.963	0.903	22.54	18.01
21	96	0.977	0.951	0.887	25.44	18.76
22	96	0.985	0.970	0.903	17.36	13.16

Frozen ensemble, 2016–2023 test data. Target: naturalized WaterGAP, not GRACE. Errors are mm; anomaly NSE removes the training monthly climatology. Higher NSE and lower errors are better.

GWSA climate reconstruction performance (01–11)

Basin	n	r	NSE	Anom NSE	RMSE	MAE
01	96	0.982	0.963	0.830	5.74	4.47
02	96	0.986	0.956	0.901	4.98	4.08
03	96	0.982	0.953	0.781	10.59	7.70
04	96	0.969	0.933	0.755	7.04	5.33
05	96	0.963	0.927	0.678	8.87	6.32
06	96	0.980	0.958	0.842	3.94	2.92
07	96	0.977	0.948	0.824	4.83	3.96
08	96	0.984	0.963	0.865	5.52	4.22
09	96	0.982	0.958	0.832	6.04	4.78
10	96	0.968	0.935	0.661	4.90	4.09
11	96	0.958	0.914	0.627	7.09	5.40

Frozen ensemble, 2016–2023 test data. Target: naturalized WaterGAP, not GRACE. Errors are mm; anomaly NSE removes the training monthly climatology. Higher NSE and lower errors are better.

GWSA climate reconstruction performance (12–22)

Basin	n	r	NSE	Anom NSE	RMSE	MAE
12	96	0.981	0.962	0.812	4.62	3.61
13	96	0.959	0.920	0.703	5.50	4.45
14	96	0.985	0.970	0.812	4.39	3.12
15	96	0.963	0.902	0.360	10.51	8.46
16	96	0.970	0.933	0.564	10.88	8.87
17	96	0.980	0.952	0.589	11.67	9.60
18	96	0.958	0.907	0.711	6.84	5.11
19	96	0.968	0.926	0.823	10.40	8.46
20	96	0.946	0.892	0.668	11.69	9.70
21	96	0.953	0.904	0.808	9.19	7.33
22	96	0.967	0.922	0.780	11.32	8.94

Frozen ensemble, 2016–2023 test data. Target: naturalized WaterGAP, not GRACE. Errors are mm; anomaly NSE removes the training monthly climatology. Higher NSE and lower errors are better.